

Input Layer

Power electronic converter



Design goals

Constraints

Improve efficiency

Ripple

reduce size

.....

Voltage

Current

Cost

.....

Design Layer

Topology Exploration

Graph Neural Networks

Topological Difference
Variational Autoencoder

.....

Parameter Optimization

Switching frequency

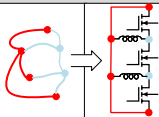
Inductance
capacitance

.....

DRL Agent

Output Layer

Topology Generation



Parameter Optimization

				
f_{switch}	L	C	V_{in}	V_{out}

N↑

Evaluate Layer

Goal 1

Goal 2

Goal 3

Goal 4

Goal 5

...

Goal n

...

Y